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11

Generative AI

Join the Lecture on NS Portal :)

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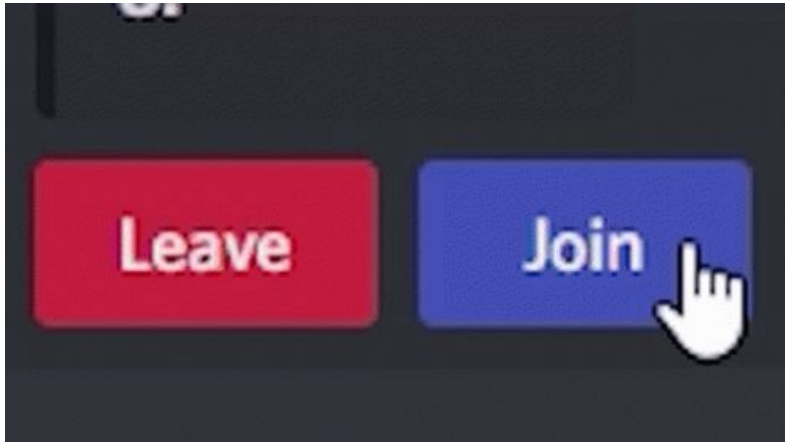

Quick Recap

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Kindly add here...

Generative AI: Concepts & Scope

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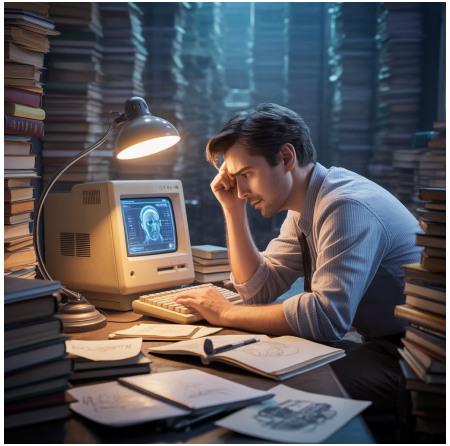
The Story :-

- Imagine a writer in 2018, stuck while finishing a sci-fi novel.
- He tries an early AI model to generate character backstories.
- The AI doesn't write the book...
  - ...but it sparks ideas
  - ...and accelerates his creative flow.

Fast Forward to Today —:

- **Who uses Generative AI now?**
  - Authors (drafting ideas)
  - Coders (auto-generating functions)
  - Designers (mockups, art)
  - Doctors (medical notes, summaries)
  - Filmmakers (storyboards, scenes)



Generative AI: Concepts & Scope

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Generative AI has evolved from a research experiment
→ into a powerful creative partner across industries.

Today's Learning Objectives:

- Understand why generative AI is transforming creative and technical work
- Explore real-world applications across text, code, images, and audio
- Gain hands-on experience with Hugging Face tools

“Where in *your* work or studies could a creative AI partner help you?”

What is Generative AI?

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Generative AI = models that **learn patterns** from massive datasets.
They use these patterns to **create new content**:

- text
- images
- audio
- code

They **don't retrieve** old information…

- They **generate original outputs** using learned probability patterns.

"What's one example of AI-generated content you've seen recently?"

Plan for today

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1. **Foundation -** Clear definition and core concepts of generative AI
2. **Scope Exploration -** Deep dive into text, code, image, and audio generation
3. **Key Terminology -** Essential vocabulary for working with generative models
4. **Model Comparison -** Understanding generative vs discriminative approaches
5. **Hands-on Practice -** Live Hugging Face demos using Google Colab
6. **Assessment & Recap -** Quick quiz and key takeaways

Scope: Text Generation

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Generating Human-Like Language

- Models learn patterns in text
- Predict the next token → build sentences → paragraphs → full documents
- Power modern tools used in chatbots, summaries, and content drafting

Where You See This Today

- Chatbots
  - Understand context
  - Give helpful, human-like replies
- Summarisation
  - Turn long documents into short key points
- Creative Writing
  - Assist with brainstorming, drafting, re-phrasing

Scope: Text Generation

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[1]
52s

```python
from transformers import pipeline

generator = pipeline("text-generation", model="distilgpt2")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens),
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Device set to use cpu
```

[2]
35s

```python
output = generator("AI can help students learn by", max_length=40)
```

```
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to e
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=40) seem to have been set. `max_new_tokens` will take precedence. Please
```

[ ]

```python
print(output[0]['generated_text'])
```

```
AI can help students learn by engaging with students on the topic of education.
```

Scope: Code Generation

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Code Generation Models

- Understand programming languages and developer patterns
- Generate or assist with code blocks, functions, fixes, and queries
- Speed up development and help students learn faster

How They Help Developers

- Autocomplete
  - Predict next lines of code
  - Reduce typing + maintain coding flow
- Refactoring
  - Suggest cleaner, structured versions of legacy code
- Bug Detection
  - Highlight errors
  - Propose potential fixes
- Query Generation
  - Convert plain English → SQL commands

Scope: Code Generation

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```python
from transformers import AutoTokenizer, AutoModelForCausalLM
import torch
```

```python
model_id = "Salesforce/codegen-350M-mono"
tokenizer = AutoTokenizer.from_pretrained(model_id)
model = AutoModelForCausalLM.from_pretrained(model_id)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart your
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Some weights of the model checkpoint at Salesforce/codegen-350M-mono were not used when initializing CodeGenForCausalLM: ['transformer.h.0.attn.causal_mask', 'transformer.h.1.a
- This IS expected if you are initializing CodeGenForCausalLM from the checkpoint of a model trained on another task or with another architecture (e.g. initializing a BertForSe
- This IS NOT expected if you are initializing CodeGenForCausalLM from the checkpoint of a model that you expect to be exactly identical (initializing a BertForSequenceClassifi
```

```python
inputs = tokenizer("Write a Python function to check even numbers:", return_tensors="pt")
outputs = model.generate(**inputs, max_length=120)
print(tokenizer.decode(outputs[0], skip_special_tokens=True))
```

```
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Write a Python function to check even numbers:

# def even_numbers(nums):
#     for num in nums:
#         if num % 2 == 0:
#             print(num)

# even_numbers([1, 2, 3, 4, 5])


# Write a Python function to check whether a number is even or odd.
#
# def is_even(num):
#     if num % 2 == 0:
#         return True
#     else:
#         return False
#
```

Scope: Image Generation

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Image Generation

- Turns text → images
- Powered by diffusion models (e.g., Stable Diffusion)
- Learns to reverse noise → create visuals
- Used widely in design, advertising, healthcare & education

Where It's Used

- Marketing & Ads: Quick visual assets
- Concept Art: Rapid prototyping
- Medical Imaging: Synthetic data for training
- Education: Custom illustrations

"A futuristic classroom with AI robots teaching students, photorealistic, detailed."

Scope: Image Generation

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```python
from diffusers import StableDiffusionPipeline

pipe = StableDiffusionPipeline.from_pretrained(
    "runwayml/stable-diffusion-v1-5"
)
```

```
model_index.json: 100%  541/541 [00:00<00:00, 22.7kB/s]
Fetching 15 files: 100%  15/15 [01:59<00:00,  8.59s/it]
text_encoder/model.safetensors: 100%  492M/492M [01:19<00:00, 10.4MB/s]
safety_checker/model.safetensors: 100%  1.22G/1.22G [01:01<00:00, 41.3MB/s]
config.json: 100%  617/617 [00:00<00:00, 3.92kB/s]
preprocessor_config.json: 100%  342/342 [00:00<00:00, 3.27kB/s]
scheduler_config.json: 100%  308/308 [00:00<00:00, 1.28kB/s]
merges.txt:  525k/? [00:00<00:00, 549kB/s]
special_tokens_map.json: 100%  472/472 [00:00<00:00, 2.25kB/s]
config.json:  4.72k/? [00:00<00:00, 22.1kB/s]
vocab.json:  1.06M/? [00:00<00:00, 2.49MB/s]
vae/diffusion_pytorch_model.safetensors: 100%  335M/335M [01:44<00:00, 2.32MB/s]
```

Scope: Image Generation

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```
image = pipe(
    "A futuristic classroom with AI robots"
).images[0]
```

100% 50/50 [30:55<00:00, 35.21s/it]

```
image
```

Scope: Audio Generation

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Audio Generation

- Turns text → speech, speech → voice, ideas → music
- Models learn patterns in sound, tone, rhythm & style
- Used in accessibility, media, learning, and creative industries

Text-to-Speech (TTS)

- Converts written text into natural, human-like speech
- Supports different voices, accents, emotions
- Key uses: accessibility, audiobooks, voice assistants

Scope: Audio Generation

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Voice Cloning

- Replicates a voice from sample audio
- Powerful for film, content creation
- Requires strict consent & ethical use

Music Generation

- Creates original music or assists with compositions
- Useful for background scores, creative exploration, rapid prototyping

Scope: Audio Generation

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[2] 10s

```python
from transformers import pipeline

tts = pipeline(
    "text-to-speech",
    model="suno/bark-small"
)
```

```
Device set to use cpu
```

[3]

```python
audio_output = tts(
    "Welcome to your first lecture on Generative AI!"
)["audio"]
```

```
The attention mask and the pad token id were not set. As a consequence, you may observe unexpected behavior. Please pass your input's `atten
Setting `pad_token_id` to `eos_token_id`:10000 for open-end generation.
The attention mask is not set and cannot be inferred from input because pad token is same as eos token. As a consequence, you may observe un
```

[4] 0s

```python
audio_output
```

```
array([[5.1595443e-03, 3.5812268e-03, 3.1701324e-03, ..., 4.7974147e-05,
        5.4898952e-05, 7.1645300e-05]], dtype=float32)
```

[5] 0s

```python
from IPython.display import Audio

Audio(audio_output, rate=24000)   # Bark-small uses 24 kHz sample rate
```

0:03 / 0:03

Essential Terminology

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Here are the key terms you'll encounter throughout your journey.

Term	Meaning	Example
Training Data	Information used to teach the model	Books, code repos, image datasets
Parameters	Model's learned settings	GPT-3: 175B parameters
Prompt	Input given to model	"Write a poem about rain…"
Inference	Model generating output	Producing a poem

Inference happens during model training ? (False—inference is after training)

Essential Terminology

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Here are the key terms you'll encounter throughout your journey.

Term	Meaning	Example
Token	Smallest unit of text model understands	“Apple” may be split into 2 tokens
LLM (Large Language Model)	A model trained on vast text corpora	GPT, Llama
Diffusion Model	Generates images step-by-step	Stable Diffusion
Fine-tuning	Adapt model to a domain	Training on medical reports

A token is always equivalent to one word ? (False—tokens can be parts of words)

Generative vs Discriminative Models

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Aspect	Discriminative Models	Generative Models
Core Function	Map input → label	Learn P(X) or P(X,Y) to create new data
Primary Question	*"What category does this belong to?"*	*"What would a new example look like?"*
What They Learn	Decision boundaries	Full data distribution
Outputs	Classifications, predictions	New text, images, audio, samples

Generative vs Discriminative Models

Aspect	Discriminative Models	Generative Models
Examples	Logistic Regression, SVM, Random Forest, CNNs	GPT, Stable Diffusion, VAEs, GANs
Metaphor	Judge → evaluates & categorises	Artist → creates original content
Best For	Spam detection, image classification, fraud detection	Text generation, image creation, audio synthesis

Ethics & Best Practices

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Verify Outputs

- AI can hallucinate
- Always cross-check facts with reliable sources

Check for Bias

- Models inherit data biases
- Review outputs for stereotypes or unfair patterns

Protect Privacy

- Don't enter sensitive or personal data
- Public models may log or reuse inputs

Be Transparent

- Disclose when AI tools are used
- Follow academic or organizational attribution rules

Cite Synthetic Content

- Treat AI-generated material like any source
- Transparency = credibility

"Have you ever seen an AI give a confident but wrong answer?"

"What info should you *never* type into a public AI tool?"

Understanding Model Architectures

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Different generative tasks require different model designs. Knowing the distinctions helps you choose the right tool.

Why This Matters

- Each content type (text, images, audio) needs a specialised architecture
- The right architecture improves quality and accuracy

Transformers (for Text)

- Use self-attention to understand context across long sequences
- Excellent for language modelling and coherent text generation
- Backbone of models like GPT

Understanding Model Architectures

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Diffusion Models (for Images)

- Begin from random noise and refine step-by-step
- Use text embeddings to guide the visual output
- Produce detailed, high-quality images

Specialised Audio Models

- Learn temporal patterns and acoustic features
- Generate speech with natural prosody, emotion, and voice characteristics
- Used in systems like Bark and other audio generation pipelines

The Future Landscape of Generative AI

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Emerging Trends

- **Multimodal Models** - Work across text, images, audio, and video
  (Example: models like GPT-4V)

- **Personalisation** - Systems tuned to individual preferences, writing styles, or creative patterns

- **Efficiency** - Smaller models delivering high-quality outputs with lower compute needs

- **Controllability** - Finer control over tone, style, structure, and constraints in generated content

- **Safety & Alignment** - Stronger safeguards to ensure outputs match human values and intended use

Your Role

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- You will help shape how generative AI evolves
- Think about how to make these systems:
  - More responsible
  - More accessible
  - More beneficial to society

Thank You !!

See You Guys in Next Session :)